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An Adaptive Cybersecurity Framework Integrating Machine Learning, Zero Trust Policy, and Blockchain for Academic Cloud Environments

Danang[1], Corresponding Author[1],*, Teguh Wahyono[2], Irwan Sembiring[3], Theophilus Wellem[4]

1,2,3,4 Faculty of Information Technology, Satya Wacana Christian University, Salatiga, Indonesia

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ABSTRACT

Cloud-based academic environments such as Learning Management Systems (LMS), Open Journal Systems (OJS), institutional repositories, and web applications face increasing cybersecurity challenges due to heterogeneous users, distributed services, and extensive exposure to public networks. Existing security approaches remain fragmented, where machine learning focuses on threat detection, Zero Trust Architecture (ZTA) emphasizes access control, and blockchain is primarily used for secure logging. The lack of integration among these components limits the ability of security systems to adapt dynamically to evolving cyber threats. This study proposes an Adaptive Cybersecurity Framework (ACF) that integrates unsupervised machine learning-based anomaly detection, a risk-based Zero Trust Policy Engine, and blockchain-based immutable audit logging within a continuous adaptive feedback loop. The framework was evaluated using 450,000 anonymized HTTP and Web Application Firewall (WAF) events collected from a multi-domain academic cloud environment consisting of LMS, OJS, repositories, and supporting web applications. The analysis revealed structured and repetitive attack behaviors dominated by automated endpoint probing and cross-domain propagation patterns, indicating ecosystem-level security threats. The proposed risk assessment mechanism demonstrated effective alignment between anomaly detection and policy-based decision making. Experimental results achieved an AUROC of 0.7296 for risk-based threat detection while maintaining an average decision latency of approximately 11 ms, indicating suitability for real-time deployment. Blockchain integration further provided verifiable, tamper-resistant audit trails for mitigation actions and policy enforcement activities. This study contributes an ecosystem-aware adaptive cybersecurity paradigm that bridges threat detection, policy enforcement, and auditability through a unified security architecture for Academic Cloud Environments.

1. Introduction

The use of cloud computing services in the higher education sector continues to increase in line with the need for campus digitalization, including in Learning Management Systems (LMS), Open Journal Systems (OJS), research repositories, and academic administration services. This digital transformation improves operational efficiency and cross-institutional collaboration, but simultaneously expands the attack surface and the complexity of security management (BSSN, 2023). Modern cloud environments face an exponential increase in web-based attacks and exploits of open public services (Iqbal et al., 2016; Khoda Parast et al., 2022; Singh et al., 2016).

Learning Management Systems (LMS) and Open Journal Systems (OJS) are among the most widely deployed academic cloud services and are frequently exposed to public networks, making them attractive targets for automated attacks. The increasing interconnectivity of these services expands the cyberattack surface and increases the complexity of security threats in academic environments. Conventional security approaches, which rely on rule-based detection, are limited in detecting dynamic, distributed, and previously unknown attack patterns. Therefore, an adaptive cybersecurity framework capable of detecting anomalous behavior, adaptive risk assessment, and contextual and dynamic security is required in the Academic Cloud Environment.

Despite significant advances in cybersecurity technology, existing approaches remain largely fragmented. Machine Learning (ML) techniques have demonstrated strong capabilities in detecting anomalies and zero-day attacks; however, these techniques often operate as standalone detection systems without direct integration into access control mechanisms (Abdalla & Jumaa, 2022; Achour et al., 2025; Adila et al., 2024).

In contrast, Zero Trust Architecture offers a modern security approach through the principles of never trust, always verify, continuous authentication, and risk-based access control (Rose et al., 2020). However, most ZTA implementations are still oriented towards access policy enforcement and have not fully integrated the real-time behavioral analysis-based adaptive intelligence needed to address increasingly dynamic and complex cyber threats (He et al., 2022).

The lack of integration between these three paradigms creates a critical gap in modern cybersecurity architecture. Specifically, detection (ML), enforcement (ZTA), and accountability (blockchain) are typically implemented separately, preventing the creation of a unified, adaptive defense mechanism. This fragmentation limits the system's ability to respond dynamically to evolving threats, especially in complex and distributed environments like academic cloud ecosystems.

Therefore, integrating ML, ZTA, and blockchain into a unified adaptive framework is crucial for achieving end-to-end cybersecurity capabilities. Such integration enables not only improved threat detection accuracy but also dynamic policy enforcement and verifiable audit trails, forming a closed adaptive security system.

The academic environment is considered high risk due to the high variety of users, cross-device access (BYOD), and weak adaptive authentication controls in some institutions (Chandramouli & Butcher, 2023). Unlike relatively homogeneous enterprise environments, academic cloud ecosystems are fragmented and distributed. A single institution may manage multiple OJS or LMS domains with varying configurations. This uniformity of endpoint structures across domains has the potential to create systemic patterns of automated reconnaissance and opportunistic exploitation.

In this study, the Academic Cloud Environment refers to interconnected academic digital services implemented in cloud-based infrastructure, including Learning Management Systems (LMS), academic information systems, research repositories, digital libraries, virtual classroom platforms, cloud storage services, and institutional web applications. The increasing interconnectivity between these services expands the attack surface and poses complex cross-domain security challenges that require adaptive cybersecurity mechanisms.

In cloud security literature, Machine Learning has become the dominant approach for anomaly detection and zero-day attack identification (Alam et al., 2025; Berrios Vasquez et al., 2025). Unsupervised models such as Isolation Forest and Deep Autoencoder have proven effective in identifying behavioral anomalies in large-scale logs without relying on static signatures.(Chen et al., 2024; FT Liu et al., 2008; Sakurada & Yairi, 2014). Deep representation learning approaches can increase sensitivity to new attack patterns in distributed systems (Javaid et al., 2016; Shone et al., 2018; Yin et al., 2017).

However, most of this research focuses on improving detection accuracy and has not yet integrated detection results directly into adaptive policy mechanisms. Detection and enforcement are often treated as two independent processes, thus not forming a continuous adaptive feedback loop.

Zero Trust Architecture (ZTA) is evolving as a modern security paradigm that emphasizes the principles of never trust, always verify, least privilege access, and continuous authentication, especially in cloud environments and distributed systems (Rose et al., 2020; Samir N. Ajani, 2024). Recent literature studies show that ZTA offers a more adaptive approach than traditional perimeter-based security, but its implementation still faces challenges in integration with Machine Learning-based detection mechanisms (Buck et al., 2021; He et al., 2022).

Attribute-based access control and context-aware policy approaches can improve the resilience of distributed systems to credential abuse and lateral movement (Hu et al., 2014; Islam & Madria, 2024). However, the ZTA literature is generally oriented towards identity-centric enforcement, while in public web services such as OJS and LMS, the dominant threats often come from automated endpoint probing and exploitation of deterministic path structures.

In addition to ML and ZTA, Blockchain has been proposed as a tamper-proof logging mechanism to improve the integrity and accountability of security systems (Hu et al., 2014; Islam & Madria, 2024; Rane et al., 2025). Blockchain-based secure logging architecture enables immutable incident recording without reliance on a single entity (Ali et al., 2022). However, blockchain integration in ML-based adaptive mitigation cycles and Zero Trust is still limited to conceptual approaches and has not been widely empirically evaluated in the academic cloud context.

Based on the literature review, three main gaps remain. First, security approaches remain fragmented, as machine learning, zero trust, and blockchain are developed separately without integration into a unified, adaptive architecture. Second, most studies evaluate security on a single domain or isolated network, without considering cross-domain propagation within homogeneous application ecosystems such as OJS or WordPress. Third, there is no explicit separation between detection-oriented anomaly scoring and policy-oriented risk scoring, resulting in a conflation between detection optimization and policy calibration.

To address these gaps, this study proposes an Adaptive Cybersecurity Framework (ACF) that integrates Machine Learning, a Zero Trust Policy Engine, and Blockchain-based immutable logging through a continuous adaptive feedback loop. Unlike existing approaches that treat detection, policy enforcement, and auditability as separate functions, the proposed framework establishes a unified

adaptive security architecture capable of dynamically responding to evolving cyber threats in Academic Cloud Environments.

The framework implements behavioral anomaly detection using Isolation Forest and Deep Autoencoder, whose outputs are incorporated into an adaptive Zero Trust risk scoring mechanism for contextual policy enforcement. In addition, hysteresis-based policy calibration is employed to improve decision stability, while blockchain immutable logging ensures the integrity and accountability of security operations.

The main novelty of this research lies in three aspects. First, it introduces a functional separation between anomaly-oriented detection and policy-oriented risk scoring, allowing threat detection and security decision-making to be optimized independently. Second, it implements an adaptive feedback loop that continuously adjusts policy thresholds based on observed threat behavior and changing security contexts. Third, it extends cybersecurity from a domain-centric defense model to an ecosystem-aware adaptive defense by incorporating cross-domain attack propagation patterns within Academic Cloud Environments.

2. Literature Review

2.1 Machine Learning-Based Anomaly Detection

Machine learning (ML)-based anomaly detection has become a leading approach to identifying unusual activity in cloud computing systems. ML can analyze large volumes of log patterns, recognize normal behavior, and detect potential anomalies without relying on static, rule-based rules that are vulnerable to zero-day attacks (Ud Din et al., 2024).

ML models such as Random Forest, Support Vector Machine, Logistic Regression, and Neural Network have been used to detect various threats, including brute force attacks, unauthorized access, SQL injection, and network traffic anomalies. Research by Alam et al. (2025) shows that a hybrid ML approach combining supervised and unsupervised learning improves detection accuracy, especially on highly variable log datasets such as those in academic cloud environments.

In addition, several studies emphasize the importance of the feature extraction process from activity logs, for example, features such as login attempt frequency, session duration, access role changes, and HTTP request patterns, which have been proven to improve model performance in identifying attacks (C. Liu et al., 2024; Qawasmeh & AlQahtani, 2025).

However, even though ML is effective in detecting threats, previous studies still have limitations, namely:

i. There is no direct integration between ML detection and access policy systems such as Zero Trust Architecture.

ii. The detection output is passive, so it does not trigger automatic mitigation or adaptive policy updates.

iii. Lack of feedback mechanisms, so the system does not learn from previous incidents to improve detection and authorization rules.

Thus, while ML plays a crucial role in anomaly detection, its capacity as part of an adaptive security system remains limited. This gap underpins the integration of ML into the Adaptive Cybersecurity Framework (ACF) proposed in this research, where ML detection results will be used to trigger Zero Trust policy updates and record mitigation actions through a Blockchain Smart Contract.

2.2 Zero Trust Architecture (ZTA)

Zero Trust Architecture (ZTA) is a modern security approach based on the principle that no entity, be it users, devices, or applications, can be trusted by default, even within an internal network. This concept is formulated in the Zero Trust Framework guidelines by NIST SP 800-207, which emphasizes continuous verification, least privilege access, and assume breach mechanisms as the main foundation (Chandramouli & Butcher, 2023).

ZTA evolved from the needs of increasingly complex and dynamic digital environments, including the rise of personal device (BYOD) usage, remote access, and distributed cloud architectures. Previous research has shown that implementing Zero Trust can reduce the risks of credential theft, lateral movement, and unauthorized access to cloud infrastructure (Kindervag et al., 2016). The Zero Trust model has also been adopted by Google through its Beyond Corp architecture, which emphasizes real-time identity, device, and security posture-based access verification (Ward & Beyer, 2014).

In the context of academic cloud security, ZTA plays a critical role because:

i. Users are very diverse (lecturers, students, educational staff, guests).

ii. Access comes from multiple devices and locations.

iii. There has been an increase in identity-based attacks such as phishing and unauthorized access.

However, existing research is still limited to static Zero Trust implementations, lacking the ability to automatically adapt to new anomalies. This is a gap in the literature that this research aims to fill by integrating ZTA with Machine Learning and Blockchain within the Adaptive Cybersecurity Framework.

2.3 Blockchain for Cybersecurity

Although blockchain is not used as a detection component, its role is crucial in ensuring the integrity of the adaptive feedback loop. The immutability of recorded policy decisions allows for verifiable tracking of policy evolution over time. In adaptive cybersecurity systems, where policies are constantly evolving, the reliability of historical decision data is crucial. Blockchain serves as a trust anchor that guarantees the integrity and non-repudiation of these records, preventing manipulation that could compromise the feedback mechanism. Therefore, blockchain is not just a record-keeping mechanism, but a fundamental component that ensures the trustworthiness of adaptive security processes.

In the proposed framework, blockchain is not used as a threat detection mechanism, but rather as a trusted security logging layer that provides three key security properties, namely immutable logging, accountability, and non-repudiation.

Immutable logging ensures that all records of mitigations, policy changes, and security events cannot be modified once recorded on the blockchain. Accountability allows all security actions to be

transparently traced back to the system decisions taken. Meanwhile, non-repudiation ensures that no entity can deny recorded security actions because all logs are cryptographically validated and permanently stored on a distributed ledger.

In the context of information security, blockchain can provide tamper-proof auditing, which means that security logs cannot be changed once recorded. This makes it highly relevant for systems requiring high data integrity and verifiable audit trails ((Sethi & Verma, 2025).

In modern security implementations, smart contracts are used to automate validation processes, policy execution, and the recording of mitigation actions without human intervention. Several studies have shown that smart contracts can reduce the risk of log manipulation, increase accountability, and strengthen incident response because every security action is recorded as a transaction validated by the blockchain network (Yang et al., 2022).

Blockchain has been widely adopted as a trusted and immutable logging mechanism for cybersecurity applications. Its ability to provide tamper-resistant records, accountability, and non-repudiation makes it suitable for recording security events and mitigation actions. However, existing studies primarily employ blockchain as an isolated auditing mechanism and rarely integrate it with adaptive anomaly detection and Zero Trust policy enforcement within a unified security architecture.

Despite its many benefits, integrating blockchain with modern security systems still poses a number of challenges, particularly regarding transaction latency, scalability, resource consumption, and the need for synchronization with other components such as Machine Learning and Zero Trust Policy Engine.

To date, most research has focused on using blockchain for security auditing or identity management in isolation. Few studies have integrated blockchain with ML-based anomaly detection and adaptive access policies (ZTA) in a single, unified architecture. Therefore, integrating blockchain into the Adaptive Cybersecurity Framework (ACF) is a significant contribution to addressing this research gap.

2.4 Synthesis and Research Gaps

A literature review revealed that there is no integrated model that simultaneously combines ML, ZTA, and Blockchain for academic cloud security. Furthermore, no framework that enables an adaptive feedback loop capable of continuously improving security policies was found. This gap formed the basis for the development of the ACF.

Based on a review of research related to machine learning for anomaly detection, Zero Trust Architecture (ZTA), and blockchain technology for security auditing, it can be concluded that each approach has demonstrated effectiveness in specific contexts. Machine learning has been shown to improve the accuracy of threat detection through automated log pattern analysis (Pawar et al., 2025; Ud Din et al., 2024). ZTA strengthens authentication and authorization processes through the principles of never trust, always verify, and continuous identity verification.(Diaz et al., 2024). Meanwhile, Blockchain enables immutable logging, thus supporting transparent and reliable security audits (Ali et al., 2022; Karaduman & Gülhas, 2025). However, the results of the literature synthesis show three major gaps:

i. There is no complete integration between ML, ZTA, and Blockchain in one adaptive architecture. Previous research has only addressed these three technologies separately. No model has yet been able to integrate them into a security system capable of automatically adapting to changing threat patterns.

ii. The absence of an adaptive feedback loop mechanism in previous research. Most studies focus only on detection (ML), or access control (ZTA), or passive auditing (Blockchain). Previous research has not combined these three components into an adaptive cycle that allows for independent policy updates based on anomaly detection results and mitigation logging.

iii. There is a lack of research focused on the academic cloud context in Indonesia.

Report (BSSN, 2023) shows that the higher education sector is one of the most common targets of attacks due to the high variety of users and weak access protection. However, there is no architectural-level adaptive security model specifically designed for academic cloud environments, which have unique characteristics such as user heterogeneity, multi-device access, and diverse digital services (LMS, OJS, research repositories).

These findings indicate that existing studies largely address detection, enforcement, and auditability as separate components. However, the lack of integration between ML-based detection, Zero Trust enforcement, and blockchain-based auditing limits the ability of current systems to respond adaptively to dynamic and distributed threats. This gap highlights the need for a unified and adaptive cybersecurity framework that can bridge these domains within a single operational architecture.

3. Methodology

3.1 Research Design

This study uses a Design Science Research (DSR) approach combined with experimental evaluation to design and validate the Adaptive Cybersecurity Framework (ACF) in an academic cloud environment. The DSR approach was chosen because this study focuses on developing an artifact in the form of a functional and applicable cybersecurity framework, and requires systematic and measurable performance evaluation. Figure 1 presents the 6-step DSR process.

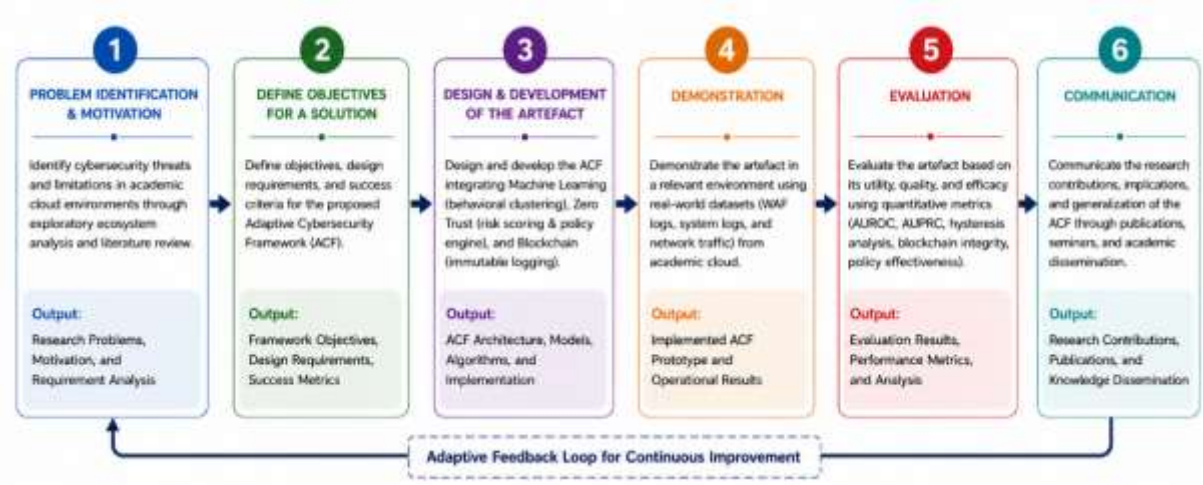


Fig. 1. Design Science Research (DSR) Process

In contrast to traditional security approaches that are signature-based and static, this research emphasizes adaptive threat detection and mitigation, where security policies can be dynamically updated based on anomaly detection results and previous incident history.

3.2 Proposed Adaptive Cybersecurity Framework

The adaptive capability of ACF is realized through a continuous feedback loop connecting anomaly detection, risk assessment, policy enforcement, and blockchain-based auditing.

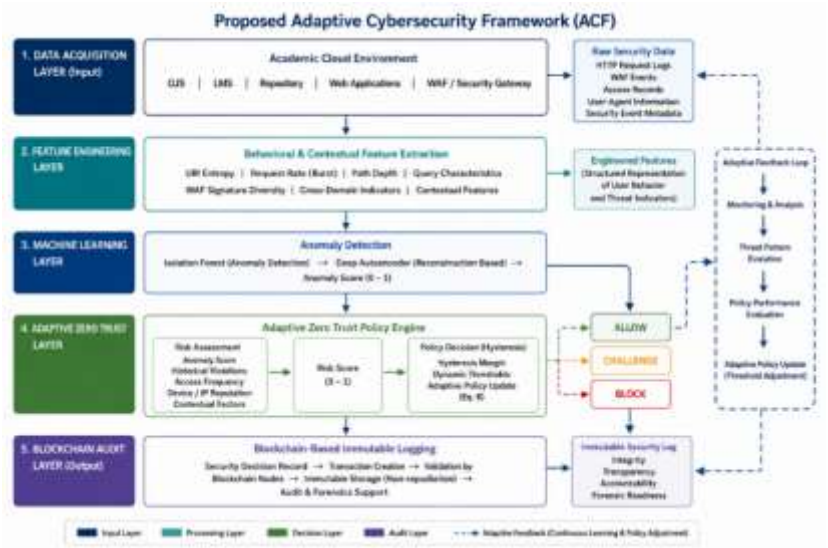


Fig. 2. Proposed Adaptive Cybersecurity Framework (ACF) Architecture

3.3 Experimental Environment

Experiments were conducted in a controlled academic cloud environment representing university information systems, such as Open Journal Systems (OJS), a Learning Management System (LMS), and a research repository. This environment was built on a virtualization-based cloud infrastructure to simulate real-world operational conditions, including user heterogeneity, multi-device access, and varying network traffic patterns.

As an initial security layer, a Web Application Firewall (WAF) was implemented to inspect incoming HTTP/HTTPS traffic. The system was then tested using controlled attack scenarios, including brute-force logins, SQL injection, unauthorized access, and traffic anomalies. These scenarios were designed to test the framework's adaptability to various types of threats common to cloud-based academic services.

The experiment was conducted on a virtual environment consisting of 3 academic service virtual machines, 1 Web Application Firewall VM, 1 Machine Learning analytics VM, and 1 permissioned blockchain node.

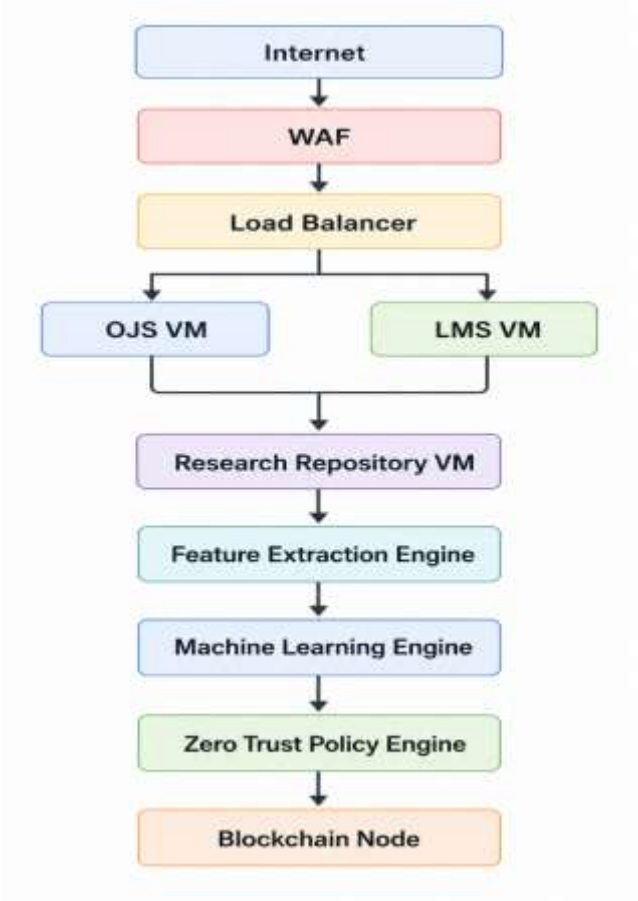


Fig. 3. Experimental Topology

3.4 Data Acquisition and Dataset Preparation

The dataset is compiled from anonymized application request logs and WAF signals (hashes for IP and identifier). To provide a clearer understanding of the dataset structure, Table 1 presents a sample of anonymized records used in this study.

Table 1. Dataset Records

Timestamp	Domain	Method	Path_norm	Status	WAF_action	URL_length	Path_depth	Request_rate_1m
2025-06-01 10:01:02	domain-01	GET	/wp-login.php	200	challenge	118	2	14
2025-06-01 10:01:03	domain-02	POST	/index.php/login	403	block	101	3	21

Each row in Table 1 represents a single HTTP request event captured from the Academic Cloud Environment. The event includes request-level attributes such as timestamp, domain, HTTP method, normalized path, response status, WAF action, URL length, path depth, and request rate within a one-minute window. The table only presents sample records to illustrate the dataset structure. The complete primary dataset consists of 450,000 security events represented by 18 engineered features derived from HTTP request logs and WAF signals.

3.4.1 Primary Dataset

The primary dataset in this study comes from security logs and access logs generated directly from an academic cloud experiment environment.

Eq. (1) represents the set of security datasets used as input in the adaptive threat modeling process in the ACF framework.

$$D = \{x_i\}_{\{i=1\}}^{N} \quad (1)$$

Where:

- D is a set of security datasets,
- (x_i) is the (i)th data observation,
- (i), is the observation index,
- N is the total number of observations in the dataset.

Each observation consists of behavioral features extracted from security logs and network traffic in the Academic Cloud Environment. This dataset is used as the primary input for machine learning, behavioral clustering, and adaptive risk assessment.

3.4.2 Secondary Dataset

For comparison and to enhance external validity, this study also utilized public datasets such as UNSW-NB15 and CICIDS2017. These datasets were used for model pretraining and as a baseline for evaluating the performance of the anomaly detection system.

3.5 Preprocessing and Feature Engineering

Raw log data is processed through data cleansing, normalization, and feature extraction. Feature engineering is performed to capture behavioral patterns and characteristics of web-based attacks, including:

i. Request rate in 1 minute and 10 minute windows (request_rate_1m, request_rate_10m).
ii. URI entropy to capture path randomness (uri_entropy).
iii. URL characteristics: URL length, query length, and path depth (url_length, query_length, path_depth).
iv. The ratio of digits and special characters in the URI for encoding/exploit indication (digit_ratio, specialchar_ratio).

v. Behavior and endpoint indicators: ua_is_bot, has_query, is_wp_path, is_admin_path, is_suspicious_ext.

Categorical features are converted to numeric form, while numeric features are normalized to maintain the stability of the Machine Learning learning process.

In this study, the extracted features not only represent the technical characteristics of HTTP requests but also form behavioral features that dynamically describe attack behavior patterns. These behavioral features include request frequency, URI entropy, path traversal characteristics, probe repetition, WAF signature diversity, and cross-domain propagation patterns.

These behavioral features are then used in an unsupervised clustering process to build adaptive threat modeling, which is a threat modeling based on actual behavioral patterns that develop in academic cloud environments. With this approach, the system not only detects known attack signatures but is also able to identify new threat patterns based on behavioral similarities and anomalous activity characteristics.

To provide a clearer description of the behavioral information used in the proposed framework, Table 2 summarizes the features extracted from the raw security logs. These features represent traffic intensity, URI structure characteristics, endpoint access behavior, and security context information derived from WAF observations. Collectively, these features form the behavioral representation used for anomaly detection, clustering, adaptive threat modeling, and risk assessment within the proposed Adaptive Cybersecurity Framework (ACF).

Table 2. Features Used in the Primary Dataset

Feature	**Description**
request_rate_1m	Number of requests observed within a 1-minute window
request_rate_10m	Number of requests observed within a 10-minute window
uri_entropy	Entropy value representing URI randomness
url_length	Length of the requested URL
query_length	Length of query string
path_depth	Number of hierarchical levels within the URI path
digit_ratio	Ratio of numeric characters in the URI
specialchar_ratio	Ratio of special characters in the URI
ua_is_bot	Indicator of bot-like User-Agent behavior
has_query	Presence of query parameters
is_wp_path	Indicator of WordPress-related endpoint access
is_admin_path	Indicator of administrative endpoint access
is_suspicious_ext	Indicator of suspicious file extensions

waf_signature_count	Number of triggered WAF signatures
waf_class_diversity	Diversity of WAF categories triggered
probe_repetition	Indicator of repeated probing activities
cross_domain_count	Number of domains accessed by the same source
burst_score	Temporal traffic burst intensity score

The final primary dataset consists of 450,000 security events represented by 18 engineered features. Each record corresponds to a single HTTP request event observed within the Academic Cloud Environment.

The output of the preprocessing and feature engineering stages forms the secondary dataset used for behavioral analysis and adaptive threat modeling. Unlike the primary dataset, which contains raw security events and request attributes, the secondary dataset consists of engineered behavioral features, normalized numerical variables, and contextual security indicators derived from the original logs. This transformed dataset serves as the primary input for clustering, anomaly detection, risk scoring, and policy adaptation processes within the proposed Adaptive Cybersecurity Framework (ACF).

Eq. (2) is used to measure the level of randomness of URI requests as an indicator of anomalous behavior in web application traffic. URI entropy is used to detect abnormal request patterns, such as encoded attacks, automated scanning, obfuscation attempts, and dynamic payload-based exploits.

$$H_{\{URI\}} = \sum P\{c \in C\}p(c)\backslash log_2 p(c) \quad (2)$$

- $H_{\{URI\}}$ = entropy of the URI distribution
- c = category/class (e.g. path, intent, etc.)
- p(c) = probability of the category appearing
- log_2 is the base two logarithm used to measure the level of information.

A higher entropy value indicates a greater degree of variation in URI characters, which generally indicates suspicious activity such as payload encoding, pattern injection, automated probing, or obfuscation attacks. Conversely, a lower entropy value indicates a more stable and normal URI pattern.

In the proposed ACF framework, URI entropy is used as one of the behavioral features in the feature engineering stage to support the behavioral clustering, anomaly analysis, and adaptive threat modeling processes in the Academic Cloud Environment. The temporal burst score is calculated using Poisson deviation.

Eq. 3 is used to measure the level of traffic activity spikes (temporal bursts) based on the deviation of the Poisson distribution. The burst score is used to identify an increase in abnormal request activity

within a certain time interval, which can indicate scanning activity, brute-force attempts, or distributed attack behavior in academic cloud environments.

$$BusrtScore = \frac{Observed - Expected}{\sqrt{Expected}} \quad (3)$$

Where:

- BurstScore is a deviation score of traffic activity,
- Observed is the actual number of requests detected in a given time interval,
- Expected is the number of requests expected based on a normal distribution or historical traffic baseline,
- $\sqrt{Expected}$is the Poisson standard deviation used to normalize traffic deviation.

A high burst score indicates a significant spike in traffic compared to normal patterns, and can therefore be an indicator of anomalous activity or an automated attack. Conversely, a low score indicates that the traffic pattern remains within the normal range of behavior.

In the proposed ACF framework, burst score is used as one of the behavioral features to support the anomaly analysis, behavioral clustering, and adaptive risk scoring processes in the Zero Trust security mechanism.

3.6 Operational Workflow of ACF

This section describes the core architecture of the proposed Adaptive Cybersecurity Framework (ACF). This framework is designed as an adaptive cybersecurity mechanism that integrates behavior-based anomaly detection, adaptive Zero Trust policy enforcement, and blockchain-based immutable logging in a single continuous feedback-driven security pipeline. The proposed architecture enables dynamic threat analysis, contextual risk assessment, adaptive mitigation, and verifiable security audits within the Academic Cloud Environment.

Operationally, ACF works through an adaptive process that begins with data acquisition and behavioral feature extraction, followed by an unsupervised clustering process to form a behavior-based adaptive threat model. The clustering and anomaly assessment results are then used by the Zero Trust Policy Engine to generate risk scores and dynamically determine mitigation decisions. All security decisions and subsequent policy changes are recorded on the blockchain as an immutable audit trail, supporting accountability and non-repudiation. See Figure 4

3.6.1 Adaptive and Integrative Mechanisms of ACF

The key contribution of the proposed Adaptive Cybersecurity Framework (ACF) lies in its adaptive and integrative design. Adaptive capabilities are realized through a continuous feedback loop connecting anomaly detection, risk assessment, policy enforcement, and audit logging. The system dynamically adjusts risk thresholds and policy parameters based on traffic patterns, historical violations, and contextual signals, enabling continuous learning and responsive security adaptation.

To ensure decision stability, hysteresis is implemented in the Zero Trust Policy Engine, reducing oscillations near thresholds while maintaining responsiveness to changes in risk levels. This mechanism balances security enforcement with user experience.

The integrative aspect of ACF is achieved through the coordinated interaction of Machine Learning (ML), Zero Trust Architecture (ZTA), and Blockchain. ML performs anomaly detection and generates risk scores, which are used by the ZTA engine for real-time access control decisions. All actions are recorded in a permissioned blockchain, ensuring an immutable and verifiable audit trail.

This integration creates a closed system where detections inform law enforcement, and law enforcement generates auditable records that support continuous improvement. As a result, ACF delivers a cohesive, adaptive, and accountable cybersecurity architecture.

3.6.2 Stages of the Proposed ACF

The proposed Adaptive Cybersecurity Framework (ACF) consists of four main stages, as illustrated in Figure 2.

i.Data Ingestion and Canonicalization

This stage collects raw security events from web application access logs, WAF logs, and optional authentication signals. The collected records undergo preprocessing, including timestamp normalization, URL canonicalization, anonymization of sensitive identifiers, and integration of WAF actions to generate a canonical event table suitable for further analysis.

ii.Feature Engineering and Interpretability Seed

At this stage, behavioral and contextual features are extracted from the canonical event records. The extracted features include URI entropy, request rate, burst score, URL characteristics, WAF signature diversity, and endpoint behavior indicators. Rule-based interpretability seeds are also generated to provide semantic references that support cluster interpretation and threat profiling.

iii.Behavior Mining and Risk Scoring

The extracted features are analyzed using unsupervised machine learning techniques, including K-Means clustering, Isolation Forest, and Deep Autoencoder. This stage aims to identify anomalous activities, discover emerging behavioral patterns, and generate anomaly scores that are transformed into risk scores for adaptive security decision-making.

Zero Trust Policy and Blockchain Audit

Risk scores generated in the previous stage are processed by the Zero Trust Policy Engine to determine security actions such as allow, challenge, or block. All security decisions and policy updates are subsequently recorded on a permissioned blockchain to provide immutable logging, accountability, auditability, and non-repudiation

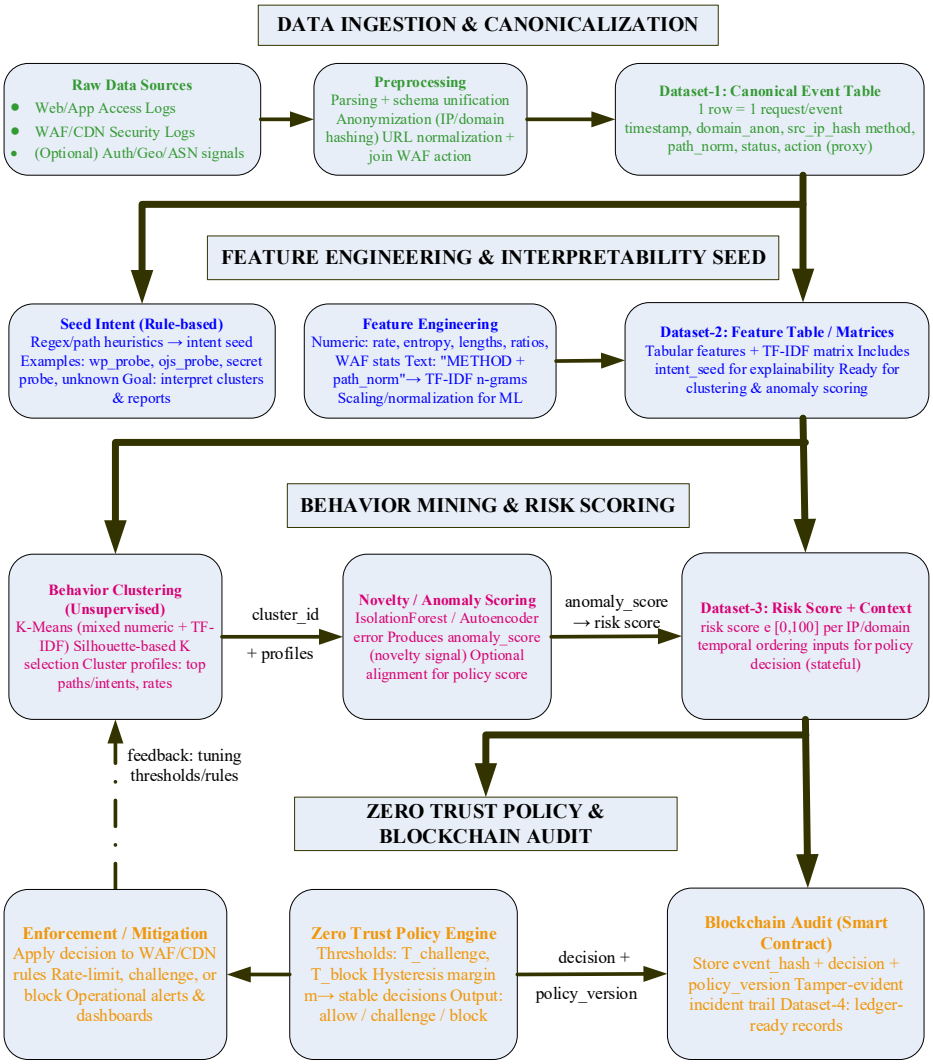


Fig. 4. Operational Workflow of the Proposed Adaptive Cybersecurity Framework (ACF)

The framework works through several integrated stages. The process begins with collecting security logs and web traffic from Academic Cloud services such as LMS, OJS, and institutional web applications. The collected data then undergoes preprocessing and feature engineering to generate behavioral features such as URI entropy, burst activity, URL characteristics, and suspicious access patterns.

These features are analyzed using machine learning models, namely Isolation Forest and Deep Autoencoder, to identify anomalous behavior and previously unknown attack patterns. The anomaly detection results are then integrated into the Zero Trust Policy Engine to generate an adaptive risk score based on behavioral factors, access history, and security context.

Based on the resulting risk score, the framework dynamically determines a security decision, whether to allow, challenge, or block. All mitigation activities and subsequent policy decisions are recorded on a permissioned blockchain to ensure immutable logging, accountability, and non-repudiation. Furthermore, the framework implements an adaptive feedback loop to continuously adjust policy thresholds based on changes in observed threat patterns.

3.7 Machine Learning-Based Anomaly Detection

The anomaly detection component uses an unsupervised machine learning approach to identify zero-day attacks. The two main models used are:

i. Isolation Forest, to detect statistical deviations in high-dimensional feature spaces. Eq. (4) is used to calculate the anomaly score in the Isolation Forest (IF) algorithm, which functions to measure the level of abnormality of an observation based on the length of the isolation path in a random isolation tree structure.

$$S_{IF}(x) = 2^{-\frac{E(h(\varkappa))}{c(n)}} \quad (4)$$

Where:

- SIF(x) is the anomaly score of observation (x),
- E(h(x))) is the expected path length to isolate observation (x),
- (h(x)) is the length of the observation isolation path on the isolation tree,
- c(n)) is a normalization factor based on the number of data samples,
- (n) is the number of observations in the dataset.

Higher anomaly scores indicate a greater likelihood of abnormal behavior, while lower scores represent normal activity patterns. In ACF, Isolation Forest supports anomaly detection and adaptive threat modeling. Although *cluster_id* and behavioral profiles are illustrated in Figure 4 as supporting inputs to the anomaly analysis stage, they are not directly used as numerical variables in the anomaly score calculation presented in Eq. (4). The anomaly score is computed exclusively from the engineered behavioral features extracted during the feature engineering stage. The *cluster_id* and behavioral profiles generated by the behavioral clustering process serve as contextual information that supports cluster interpretation, threat profiling, and adaptive threat modeling. These elements improve the explainability of detected anomalies by providing semantic descriptions of behavioral patterns observed within the Academic Cloud Environment.

ii. Deep Autoencoder, to learn the representation of normal traffic behavior and detect anomalies based on reconstruction error.

Deep Autoencoders are used to learn representations of normal network traffic behavior through the process of encoding and decoding security features. This model detects anomalies based on reconstruction error, which is the degree of difference between the original input data and the model's reconstruction.

Reconstruction error is calculated using Eq. (5):

$$S_{AE}(\varkappa = \ [\![x - \varkappa]\!]^2 \quad (5)$$

Where:

- (SAE(x) is the reconstruction error or anomaly score of observation (x),

- (x) is the original input data,
- ($\varkappa$is the result of data reconstruction by Deep Autoencoder,
- $[\![x - \varkappa]\!]^2$is the squared distance (squared reconstruction error) between the original data and the reconstructed results.

A high reconstruction error value indicates that the model failed to adequately reconstruct the data pattern, making the observation more likely to be anomalous activity or a cyberattack. Conversely, a low reconstruction error value indicates that the observed behavior pattern still aligns with the normal traffic characteristics the model has learned.

In the proposed ACF framework, Deep Autoencoder is used to support adaptive anomaly detection based on behavioral features in the Academic Cloud Environment, particularly in detecting unknown attack patterns and traffic behavior that deviates from the normal baseline.

3.8 Adaptive Zero Trust Policy Engine

In the proposed ACF architecture, the Machine Learning component does not directly determine access decisions, but instead generates an anomaly score to represent the level of abnormality in traffic behavior. The Zero Trust Policy Engine then processes this anomaly score along with other contextual parameters such as breach history, access frequency, device characteristics, and time context to generate a risk score used in security decision-making.

Thus, the risk score in this study is a policy-oriented risk assessment generated by the Zero Trust mechanism, while the anomaly score functions as a detection-oriented indicator derived from the Machine Learning model.

The Zero Trust Policy Engine serves as a risk-based access decision-making mechanism. Access decisions are determined through the following functions:

policy_decision(risk_score) = { block, if risk_score >= T_block; challenge, if T_challenge <= risk_score < T_block; allow, if risk_score < T_challenge }.

To avoid decision fluctuations (oscillations) in traffic around the threshold, the framework applies hysteresis with a margin m. In summary, the transition to a stricter risk level uses the upper threshold (*T_challenge and T_block*), while the risk level reduction uses the lower threshold (*T_challenge - m and T_block - m*).

Eq. (6) is used to calculate the dynamic risk score in the Zero Trust Policy Engine mechanism. The risk score represents the security risk level of a user entity based on a combination of anomaly detection results, access behavior, security history, and operational context in an adaptive manner.

$$R(_{U,t}) = \alpha A(x) + \beta F(_{U,t}) + \gamma H(_{U}) + \delta CR(_{t}) \quad (6)$$

Where:

- R(U,t)) is the user's risk value (U) at time (t),
- A(x) is the anomaly score generated from the Machine Learning model,

- F(U,t)) is the deviation of the user request frequency from the normal access pattern,
- H(U)) is the user's history of violations or suspicious activity,
- C(t)) is a contextual factor that includes location, device, access time, and system environmental conditions,
- $\alpha, \beta, \gamma, \delta$ is the risk parameter weight used to control the contribution of each factor to the final risk value.

Higher risk scores trigger stricter security responses, while lower scores allow normal access. The risk score serves as the primary input for adaptive Zero Trust decision-making.

Equation (7) is used to determine adaptive access decisions in the Zero Trust Policy Engine mechanism based on the risk score value generated from the adaptive risk assessment process.

$$Decision = \begin{cases} Allow & R < \theta_1 \\ Challenge & \theta_1 \leq \Theta_2 \\ Block & R \geq \Theta_2 \end{cases} \quad (7)$$

Where:

- Decision is a security decision that the system implements,
- R is the risk score value resulting from the adaptive risk evaluation,
- θ_1) is the low risk threshold,
- θ_2) is the high risk threshold,
- Allow indicates that access is permitted,
- Challenges indicate that the system implements additional verification or adaptive mitigation,
- Block indicates that access is blocked due to a high level of risk.

Access decisions are made dynamically based on the risk level of user behavior and the detected security context. If the risk value is below the low threshold (1), the activity is considered safe and access is granted normally. If the risk value is in the medium range, the system applies challenge mechanisms such as multi-factor authentication, rate limiting, or adaptive verification. Meanwhile, if the risk value exceeds the high threshold (2), the system automatically blocks access to prevent potential security threats.

In the proposed ACF framework, this adaptive access decision mechanism is the core of the Zero Trust Architecture implementation which enables contextual, dynamic, and risk-based security policy enforcement in the Academic Cloud Environment.

Equation (8) is used to adaptively update security policy parameters based on changes in the threat level detected in the system environment. This mechanism allows the Zero Trust Policy Engine to dynamically adjust risk thresholds according to actual threat conditions.

$$\theta_{t+1} = \theta_1 + \eta(ObservedThreatRate - TargetThreatRata) \quad (8)$$

Where:

- $\theta_{t+1}$is the policy threshold parameter in the next period,
- (t) is the policy threshold parameter in the current period,θ
- $\eta$is the learning rate or adaptive adjustment rate,
- (ObservedThreatRate) is the actual threat level detected by the system,
- (TargetThreatRate) is the target threat level or expected security baseline.

If the actual threat level exceeds the security target, the policy threshold will be adjusted to be stricter to increase the sensitivity of threat detection and mitigation. Conversely, if the threat level decreases, the system can adjust the threshold to avoid excessive false positives.

In the proposed ACF framework, this adaptive policy update mechanism supports continuous security adaptation in the implementation of Zero Trust Architecture, so that security policies can develop dynamically following changes in attack patterns and threat behavior in the Academic Cloud Environment.

3.9 Blockchain-Based Security Logging

To ensure integrity and accountability, every mitigation action and policy update is recorded using a Blockchain Smart Contract on a permissioned blockchain network. To maintain privacy and efficiency, the smart contract stores a minimal summary of event_hash (*hash log/even*t), *policy_decision (allow/challenge/block), policy_version, and timestamp*, while the raw log remains off-chain. This design maintains immutable and non-repudiation properties and allows for auditing of policy changes triggered by the feedback loop.

Mitigation actions and security decisions within the ACF framework are recorded on a permissioned blockchain network to ensure the integrity, auditability, and non-repudiation of all system security activities. Data stored on the blockchain is represented as a hash of events as follows on Eq. (9):

$$T_x = Hash(Event_ID||Risk||Action||Timestam) \quad (9)$$

Where:

- T_x is the security transaction hash stored on the blockchain,
- Hash is a cryptographic hash function,
- (Event_ID) is the unique identity of the security event,
- Risk is the risk score value resulting from the Zero Trust Policy Engine evaluation,
- Action is a mitigation action or security decision that the system applies,
- Timestamp is the time the security event was recorded,

Through hashing and a distributed ledger, every security record stored on the blockchain is immutable and cannot be modified without altering the integrity of the entire blockchain. Thus, the system provides a trusted audit trail that supports accountability, integrity assurance, and non-repudiation across all adaptive mitigation processes implemented by the ACF framework.

In this research, permissioned blockchain is used as a trusted logging layer to ensure that all security decisions, policy changes, and mitigation activities can be verified transparently and securely in the Academic Cloud Environment.

The performance evaluation of the ACF framework was conducted to measure the operational efficiency of the system, particularly regarding mitigation confirmation time and the computational overhead generated by the adaptive cybersecurity mechanisms.

Mitigation latency is calculated using Eq. (10):

$$Latency = T_{confirm} - T_{submit} \quad (10)$$

Where:

- (Latency) is the time delay in the mitigation process,
- (Tconfirm is the confirmation time of a security decision or blockchain recording,
- (Tsubmit is the time a security event is sent to the system.

Latency values are used to measure the framework's response speed in processing threat detection, adaptive decision making, and security logging on the blockchain.

Meanwhile, the computational overhead of the framework is calculated using Eq. (11):

$$Overhead = \frac{CPU_{ACF} - CPU_{baseline}}{CPU_{baseline}} \quad (11)$$

Where:

- Overhead is the ratio of the additional computational load generated by the ACF framework,
- CPUACF is the CPU resource usage when the ACF framework is enabled,
- CPUbaseline is the CPU usage on a baseline system without the adaptive framework.

The overhead value is used to evaluate the efficiency of the framework implementation in terms of system resource consumption. The lower the overhead value, the more efficient the implementation of the proposed adaptive cybersecurity framework.

In this study, latency and computational overhead evaluations were used to ensure that the integration of Machine Learning, Zero Trust risk scoring, and blockchain immutable logging remains capable of providing adaptive security performance without producing significant system degradation in the Academic Cloud Environment.

3.10 Evaluation Method

This evaluation is designed to validate the core objectives of the proposed Adaptive Cybersecurity Framework (ACF), specifically its integrative architecture (ML, ZTA, and blockchain) and its adaptive capabilities in responding to dynamic threat patterns.

Exploratory Data Analysis (EDA) was performed to understand the behavioral characteristics of security events, identify attack patterns, analyze domain exposure, and investigate cross-domain propagation behavior within the Academic Cloud Environment.

Beyond the Exploratory Data Analysis (EDA) findings and behavioral clustering summary (Table 5), the evaluation also includes a preliminary quantitative assessment of the risk scoring and Zero Trust Policy Engine components using proxy labels derived from the defense system (action==block). The evaluation includes AUROC/AUPRC metrics, threshold sweeps, and hysteresis margin studies to assess decision stability. For the blockchain component, we report latency and hash integrity metrics on the testbed as an indication of audit trail overhead.

Evaluation is performed at two levels: (a) analysis of seed intent and cross-domain propagation (without supervised labels), and (b) evaluation of the risk scoring component and Zero Trust Policy Engine using proxy labels from the defense system (e.g., *action==block, log_source==waf*, or *attack_type!=unknown*). The focus of evaluation (b) is the alignment of risk scores to operational policies, not the verification of successful exploitation.

To maintain operational realism, data is time-sorted per source (*src_ip_hash*) so that the hysteresis and adapted rate simulations follow the sequence of events. Additional analysis can be performed with generalization scenarios (e.g., *domain/IP holdout*) to test the model's dependence on domain identity.

For clustering, the number of clusters was selected using the silhouette metric. The cluster summary includes cluster size, block_rate, probe_rate, dominant intent, and the main path patterns that appear in each cluster.

The performance of the risk assessment component was evaluated using AUROC and AUPRC against proxy labels, including comparisons between score variances (e.g., anomaly score vs. policy score = 1 − anomaly score). These results demonstrate the effectiveness of the ML component in identifying anomalous patterns that support improved threat detection. AUROC was selected because it measures the capability of the risk assessment mechanism to discriminate between high-risk and low-risk activities across various decision thresholds. This metric is particularly suitable for cybersecurity datasets, where the distribution of normal and malicious activities is typically imbalanced. Therefore, AUROC provides a more comprehensive assessment of discrimination capability than accuracy alone.

However, the adaptive capability of the proposed framework cannot be fully represented by AUROC. To evaluate adaptiveness, additional metrics were employed, including policy decision latency, adaptation rate, threshold adjustment behavior, and oscillation reduction achieved through the

hysteresis mechanism. These metrics collectively assess the framework's ability to continuously adapt security policies in response to evolving threat patterns while maintaining decision stability.

For policy-level evaluation, the outputs (allow/challenge/block) are analyzed in terms of decision distribution, adaptation rate (i.e., decision change across sequential events per source), and precision/recall for the "block" class. The use of hysteresis is further examined through threshold testing and margin analysis, which demonstrate its role in reducing decision oscillations while maintaining responsiveness—thus confirming the framework's adaptive capabilities.

From an integration perspective, the evaluation is intended to assess machine learning (ML)-derived risk scores effectively translate into Zero Trust policy decisions, demonstrating a seamless link between the detection and enforcement layers. Furthermore, blockchain performance was evaluated using latency and hash integrity metrics in a test environment, confirming the feasibility of secure and immutable audit logging at a manageable operational cost.

Finally, the evaluation highlights the trade-off between risk appetite and operational friction. Aggressive thresholds increase recall but increase challenge/block rates, while conservative thresholds reduce friction at the expense of missed threats. Therefore, threshold testing and hysteresis calibration are used to provide practical policy recommendations.

Overall, the proposed evaluation framework is designed to assess the effectiveness of the Adaptive Cybersecurity Framework (ACF) in integrating threat detection, Zero Trust policy enforcement, and blockchain-based auditing. The evaluation focuses on measuring detection performance, policy adaptation capability, decision stability, and audit logging feasibility within Academic Cloud Environments.decisions, and ensuring verifiable security operations in academic cloud environments.

4 Results and Discussion

4.1 Exploratory Ecosystem Analysis Result

As part of the initial phase of the Design Science Research (DSR) approach, namely problem identification and motivation, an exploratory analysis of the traffic ecosystem was conducted to identify threat characteristics and the complexity of the academic cloud security environment. This analysis is not only descriptive, but also serves as an empirical foundation in formulating the design requirements of the Adaptive Cloud Framework (ACF) artifact. Through an Exploratory Data Analysis (EDA) approach, this section shows that the emerging threat patterns are structured, repetitive, and cross-contextual, thus cannot be effectively addressed by conventional static or rule-based security approaches.

4.1.1 Seed Intent Distribution

The semantic URI-based seed intent distribution in Table 3 shows that the majority of anomalous activity is concentrated in the *ojs_generic, php_shell_probe*, and *ojs_plugin_probe categories*. The dominance of these categories indicates that most attacks are opportunistic scanning attacks that exploit the uniformity of endpoint structures across the OJS and WordPress platforms.

Significant proportions in the *php_shell_probe and wp_auth_probe* categories indicate systematic initial probing attempts. This pattern is consistent with automated, bot-based campaigns targeting generic endpoints without relying on domain-specific context.

Table 3 Semantic URI-based intent seed distribution

intent_seed	**count**
ojs_generic	122599
php_shell_probe	65853
ojs_plugin_probe	60577
unknown	55687
wp_generic	39605
ojs_login	38107
wp_auth_probe	34633
crawler_probe	20428
ojs_file_download	20104
env_or_secret_probe	5204
ojs_admin_or_workflow	1145

The presence of a significant number of unknown categories demonstrates the limitations of a purely rule-based approach. This indicates that some attack patterns cannot be classified deterministically using static rules.

In the context of DSR, these findings trigger a design requirement: a data-driven approach based on behavioral clustering to adaptively expand the intent taxonomy. Thus, the distribution of seed intents is not merely descriptive but also provides empirical evidence of the gap between threat complexity and the capabilities of conventional security systems.

4.1.2 Cross-Domain Propagation

The analysis was conducted on security events collected from five virtual servers (VMs) hosting multiple academic cloud services. The observed environment consisted of 19 domains distributed across Open Journal Systems (OJS), Learning Management Systems (LMS), institutional repositories, and supporting web applications. Cross-domain propagation refers to activities originating from the same source entity (*src_ip_hash)* that interact with multiple domains or services within this environment. Such behavior may indicate automated reconnaissance, credential reuse attempts, or coordinated attack campaigns targeting multiple services simultaneously.

Table 4 and Figure 2 shows that some sources (IP hashes) interact with more than one domain and even across servers. This phenomenon indicates reuse of tools and exploitation of application configuration uniformity across institutions.

Table 4 Example sources (IP hashes) with the highest cross-domain activity (top 9).

attacker_id	events	domains	servers	first_date	last_date	ojs_like_rate	wp_like_rate	probe_rate	block_rate	top_intent_seed	top_intent_count	cross_domain_flag
IP-7e2d8dc0c901…	95	4	2	2026-01-31 03:59:31	2026-02-11 22:04:23	0.000	0.221	0.000	0.958	unknown	65	1
IP-f5409c52f87f…	125	3	1	2026-01-30 20:11:27	2026-02-02 16:34:07	0.000	0.000	0.000	1,000	unknown	125	1
IP-0c454359c225…	69	3	1	2026-02-17 09:12:16	2026-02-18 02:29:56	0.000	0.000	0.000	1,000	unknown	69	1
IP-a29b4cb8960d…	16	3	1	2026-02-10 00:50:28	2026-02-12 18:42:44	0.000	0.000	0.000	1,000	unknown	13	1
IP-4aca16c4fdec…	2376	2	1	2025-11-04 21:32:14	2025-11-30 18:45:50	0.021	0.928	0.016	0.012	php_shell_probe	1173	1

attacker_id	events	domains	servers	first_date	last_date	ojs_like_rate	wp_like_rate	probe_rate	block_rate	top_intent_seed	top_intent_count	cross_domain_flag
IP-f8b2894020 6a…	1026	2	1	2025-10-27 19:48:40	2025-11-23 21:21:55	0.059	0.612	0.000	0.019	wp_generic	582	1
IP-c50f28cfdb 64…	661	2	1	2025-10-29 08:37:52	2025-10-29 13:38:33	0.197	0.808	0.026	0.003	php_shell_probe	292	1
IP-0a926a9433 14…	123	2	1	2025-10-31 07:25:45	2025-11-01 11:07:06	0.179	0.919	0.000	0.041	php_shell_probe	69	1
IP-d503b6bbda 7f…	111	2	1	2025-10-31 09:40:4	2025-11-17 10:12:24	0.333	0.342	0.018	0.595	php_shell_probe	32	1

Cross-domain propagation findings: Some attackers operate on more than one domain/server, indicating reuse of tools and exploitation of uniform OJS/WordPress configurations across institutions. This pattern is relevant for ecosystem risks: a single automated campaign can spread rapidly to multiple domains with similar attack surfaces.

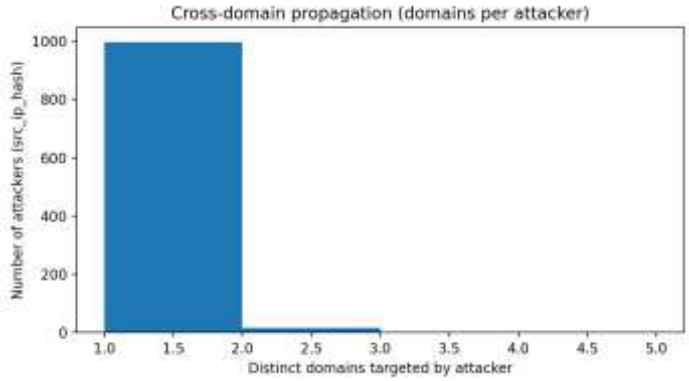


Fig. 2. Distribution of the number of domains targeted per attacker (cross-domain propagation indicator).

The distribution of the number of domains targeted per attacker shows that although the majority of sources only target one domain, there is a small group that targets multiple domains simultaneously.

From an ecosystem security perspective, this group has a systemic impact because a single automated campaign can spread rapidly across a homogeneous infrastructure. This demonstrates the inadequacy of individual domain-based mitigation approaches.

Within the DSR framework, these findings point to the need for an ecosystem-based (cross-domain awareness) security model and cross-entity correlation-based detection mechanisms, which are then accommodated in the design of the ACF artifact.

4.1.3 Domain Exposure Analysis

The per-domain exposure statistics in Table 5 and Figure 3 show significant variation in the number of attacker_ips, attacker_events, and intent distribution across domains. Some domains exhibit a high volume of anomalous interactions with a relatively low block_rate, while others have a lower volume but a block_rate close to 1.

Table 5 Exposure statistics per domain (anonymized; top 10).

domain_anon	attacker_ips	attacker_events	ojs_like_rate	wp_like_rate	block_rate
Domain-01	565	86207	0.177	0.743	0.033
Domain-02	266	45018	0.152	0.787	0.050
Domain-03	80	3062	0.791	0.031	0.070
Domain-04	85	2503	0.333	0.425	0.157
Domain-05	10	279	0.007	0.018	1,000
Domain-06	7	176	0.034	0.040	1,000
Domain-07	6	115	0.052	0.061	1,000
Domain-08	6	69	0.000	0.014	1,000

domain_anon	attacker_ips	attacker_events	ojs_like_rate	wp_like_rate	block_rate
Domain-09	4	54	0.463	0.000	1,000
Domain-10	6	42	0.000	0.000	1,000

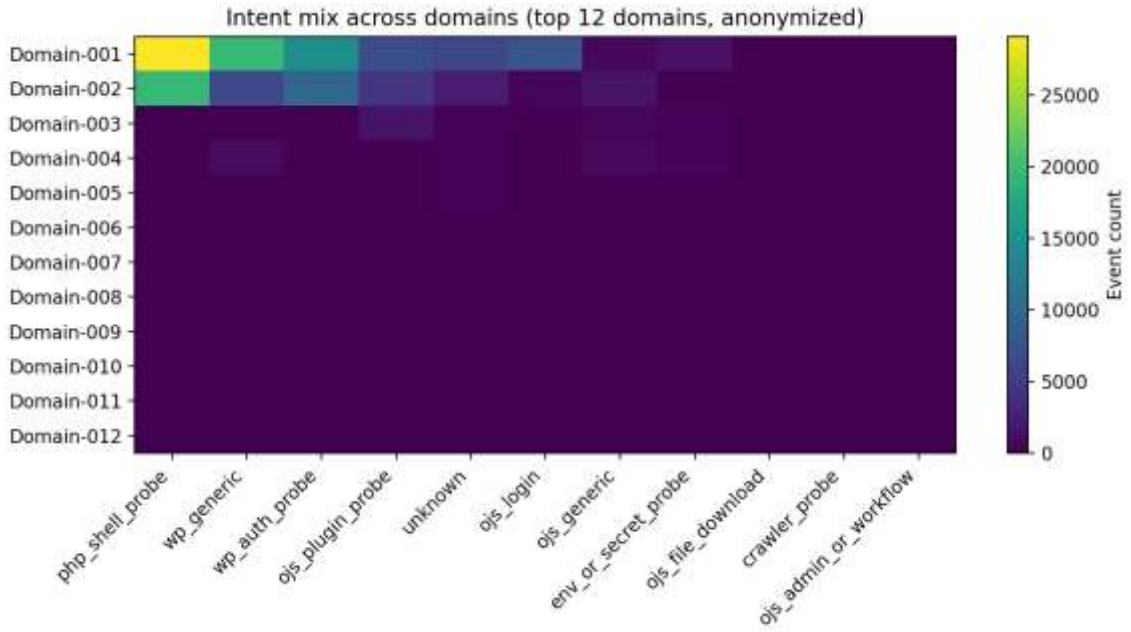


Fig. 3. Anonymized domain–intent heatmap (top 12 domains) to illustrate intent variation across domains.

This variation indicates that each domain has a unique risk profile influenced by configuration factors, exposure levels, and possibly differing levels of security hardening.

Differences in *ojs_like_rate and wp_like_rate* values across domains indicate that attack patterns are not uniform. Thus, a static threshold policy approach could potentially result in over-blocking of some domains and under-protection of others.

From a DSR perspective, these findings reinforce the need for a contextual and adaptive risk scoring mechanism that takes into account the specific characteristics of each domain, as implemented in the ACF artifact through the context-aware Zero Trust Policy Engine.

4.2 Adaptive Cybersecurity Framework Design Objectives

Based on an exploratory ecosystem analysis, the proposed Cybersecurity Framework (ACF) is designed to address the limitations of conventional rule-based security mechanisms in academic cloud environments, particularly in addressing dynamic, distributed, and previously unknown cyberthreats. The framework aims to develop an adaptive security mechanism capable of identifying evolving attack behaviors through behavior clustering and anomaly analysis derived from security logs and web traffic patterns.

Furthermore, the framework aims to provide adaptive and context-aware security enforcement by integrating a Zero Trust risk assessment mechanism that dynamically evaluates user behavior, contextual access patterns, and anomaly indicators. To strengthen the integrity and accountability of security operations, the proposed framework also incorporates blockchain-based immutable logging to establish a secure, transparent, and verifiable audit trail for mitigation activities and policy enforcement processes.

Building on the exploratory ecosystem analysis, the proposed ACF operationalizes its security objectives through behavioral clustering, which identifies recurring attack patterns, extends detection beyond static rules, and supports risk scoring and Zero Trust policy enforcement.

In response to the limitations of rule-based approaches in capturing dynamic and unknown attack patterns, the behavioral clustering component was formulated within the DSR framework with three primary objectives: (1) to identify attack behavior patterns adaptively based on data, (2) to expand the detection scope beyond static rules, and (3) to generate high-level intent representations that support Zero Trust-based risk scoring and decision-making mechanisms.
To achieve these goals, a behavioral clustering approach is used as part of the Adaptive Cloud Framework (ACF) artifact design.

Clustering was performed using K-Means on the combined representation of TF-IDF (method + normalized path) and behavioral numerical features. From the k candidates$\in$$\{5,7,9,12,15\}$, k=5 is chosen with silhouette score ≈0.3189.

Moderate silhouette values reflect the complexity and natural overlap in large-scale web traffic data. In a security context, the goal of clustering is not perfect segmentation, but rather the abstraction of dominant behavioral patterns that can be interpreted as high-level intent.

Note that additional validation metrics such as Davies–Bouldin can be reported in the clustering re-run stage; in the dataset2_final artifact exported for the policy pipeline, cluster labels are not included so recalculation of these metrics is not available without re-running the clustering stage.

Table 6 Summary of behavioral clusters (k=5; silhouette≈0.3189)

cluster	**n**	**domains**	**social studies**	**block_rate**	**probe_rate**	**ojs_like_rate**	**wp_like_rate**	**top_intent**	**top_paths**
0	79245	4	962	0.056	0.046	0.252	0.667	php_shell_probe; wp_generic; wp_auth_probe…	/index.php/index; /wp-content/plugins; /wp-content/themes; /wp-incl…
3	30080	2	56	0.015	0.038	0.058	0.896	php_shell_probe; wp_auth_probe;	/wp-content/themes; /wp-includes/blocks;

								wp_generic …	/wp-admin; /wp-login.php; …
1	9983	2	13	0.008	0.008	0.035	0.802	wp_generic; php_shell_probe; unknown; unknown; wp_a…	/wp-includes/blocks; /wp-content/themes; /administrator/components;…
2	594	13	35	1,000	0.002	0.066	0.027	unknown; unknown; php_shell_probe; ojs_generic; env…	/index.action; /index.php/index; /login.action; /.env.bak; /; /admi…
4	98	1	1	1,000	0.010	0.000	0.000	unknown; env_or_secret_probe	/register; /.env-bak; /.env-backup; /.env; /.env.bak; /.env.server;…

Table 6 shows that some clusters have a high *probe_rate* with repetitive path patterns, indicating automated scanning activity. Other clusters show a *block_rate* close to 1 with predominantly unknown intent, potentially representing exploitation patterns not yet captured in the seed taxonomy. Thus, clustering serves as an adaptive enrichment mechanism that expands the detection scope from a rule-based approach to a behavior-based one.

Within the DSR framework, the behavioral clustering component acts as an adaptive enrichment mechanism that expands detection coverage from a rule-based approach to a behavioral one, while also providing a more dynamic intent representation to support risk scoring and the Zero Trust Policy Engine. Thus, clustering is positioned not merely as a data analysis technique, but as an integral part of the ACF artifact design, enabling the system to adapt to evolving threat patterns. To provide a more comprehensive evaluation of the proposed framework, the analysis of risk scoring is extended to the Zero Trust Policy Engine. This extension is important because the risk score does not function solely as a detection output, but also serves as an operational input for adaptive access control decisions, including allow, challenge, and block actions.

As part of the fourth stage in the Design Science Research (DSR) approach; demonstration, the developed artifact in the form of the Adaptive Cloud Framework (ACF) was tested using a real dataset to demonstrate its operational feasibility in the context of academic cloud security.

The dataset used in this study is a subset curated and sampled from production logs spanning millions of events. This subset was used to maintain computational efficiency without compromising the representativeness of the analyzed attack patterns. In the evaluation subset, the "server" identity is represented by the WAF/collector node recorded as *server_id*, with a total of two observed nodes.

Meanwhile, the "domain" entity is derived from the *site_domain* attribute, with a total of 28 unique domains identified.

Conceptually, the ACF artifact is designed to operate in a broader multi-server, multi-domain environment (e.g., up to 5 servers with approximately 75 domains). However, this scope is not fully represented in this subset of evaluations, so the results obtained reflect an initial proof-of-concept rather than a full-scale implementation. Table 7 presents a summary of the dataset characteristics and proxy label statistics used in the evaluation of the risk scoring components and the implementation of the Zero Trust policy.

Table 7 Dataset overview and proxy label statistics

Metric	Value
Total events (sample used for evaluation)	450,000
Total WAF/edge nodes observed (server_id)	2
Unique domains observed (site_domain)	28
Unique IP (hashed)	75,433
Total WAF-triggered (block) events	96,244
% block (WAF)	21.388%
Observation period (Asia/Jakarta)	2025-05-30 09:32:05 – 2026-02-19 04:19:44
Sampling note	Subset (N=450,000) of million-scale production logs; used for computational efficiency and initial evaluation.

In the context of DSR, this dataset functions as a medium for:

I. Demonstrate the capabilities of the artifact (ACF) in managing and analyzing real traffic,

II. Provide an empirical basis for the evaluation stage (stage 5 of DSR),

Testing the consistency of the risk scoring mechanism against attack patterns represented through WAF-based proxy labels (block events).

Thus, this section is not only descriptive, but also an integral part of the artifact validation process in the DSR cycle, particularly in ensuring that the proposed solution can be operationalized on real-world data.

4.3 Risk Scoring Evaluation

To provide a comparative perspective, the proposed framework is evaluated against several baseline approaches, namely: (1) standalone anomaly detection using anomaly_score, (2) static threshold-based policy without adaptive feedback, and (3) rule-based WAF decision as an operational baseline. This comparison shows that while anomaly_score has stronger discriminatory capabilities, the

proposed risk_score provides better alignment with policy enforcement needs. In addition, the hysteresis-based adaptive policy mechanism is able to improve decision stability compared to the static threshold approach.

This section evaluates the performance of risk assessment as a core component in the Adaptive Cybersecurity Framework (ACF), particularly in its role of bridging anomaly detection and policy enforcement through adaptive feedback mechanisms.

It should be noted that the evaluation in this study uses proxy labels derived from WAF actions (e.g., *action == block*), rather than directly validated attack labels (ground truth). This reflects real-world conditions in operational environments, where cybersecurity datasets with complete and validated labels are often unavailable. While proxy labels do not fully represent attack success, this approach provides a practical representation of security-relevant events and is commonly used in the evaluation of security detection and response systems. Therefore, the evaluation focuses on alignment with operational mitigation decisions, rather than on absolute attack classification accuracy.

In this context, the evaluation is performed using the proxy label y=1y = 1y=1 when action == block. The primary purpose of this evaluation is to assess the score's ability to rank events leading to mitigation actions, not to verify successful exploitation.

Table 8 summarizes the discriminatory performance of two types of scores: anomaly_risk (= 1 − anomaly_score) and risk_score (*score_policy*). The anomaly_score transformation is performed to maintain the consistency of the score direction, where higher values indicate greater risk. Meanwhile, *risk_score* is a policy-aligned score and is used as input for the Zero Trust policy engine.

The results show that anomaly_risk has a higher discriminatory ability (AUROC 0.7296; AUPRC 0.3158) compared to *risk_score* (AUROC 0.5727; AUPRC 0.2227). The AUPRC value of risk_score is close to the baseline prevalence of the positive class (≈0.2139) indicating that this score plays a role more as a policy gating mechanism than as an optimal predictor for ranking events that lead to blocking. This finding emphasizes the importance of separating the functions between the detection score (*anomaly_risk*) and the policy score (*risk_score*) in the ACF architecture.

Table 8 Discriminative performance of scores (AUROC and AUPRC) against the action==block proxy label.

Metric	**anomaly_risk (1−anomaly_score)**	**risk_score (score_policy)**
AUROC	0.7296	0.5727
AUPRC	0.3158	0.2227

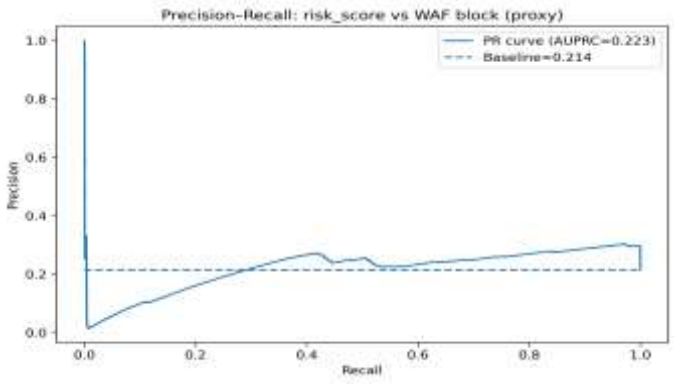


Fig. 4. Precision-Recall: risk_score vs WAF block (proxy).

Figure 4 visualizes the Precision-Recall behavior of risk_score against the proxy label action=block. The curve is close to the baseline of positive class prevalence, consistent with the AUPRC values approaching the baseline. This finding supports the interpretation that *risk_score* acts as a policy-aligned score for friction control via thresholds and hysteresis, rather than as a single optimal predictor for ranking events that lead to WAF blocking.

The score distribution shows that most events fall within the low-risk range, while the tail of the distribution contains events with a higher proportion of blocks. In policy practice, this tail is the focus for challenge/block decisions.

From a DSR perspective, this evaluation results reveal two main implications for the design of ACF artifacts. First, *anomaly_risk* is effectively used for detection prioritization and early identification of anomalous patterns. Second, risk_score serves for policy enforcement and friction management through threshold and hysteresis tuning. Thus, these two scores have complementary roles in the ACF architecture.

The evaluation results show that the risk scoring mechanism in ACF has sufficient discriminatory capacity to support adaptive security decision-making. Furthermore, the separation of functions between detection and policy scores is crucial in ensuring a balance between effective mitigation and control of system friction.

These results indicate that the proposed framework relies not only on the accuracy of anomaly detection, but also on the effective integration between detection and policy enforcement to achieve adaptive cybersecurity.

To further evaluate the operational effectiveness of the proposed framework, the generated risk_score is assessed in the Adaptive Zero Trust Policy Engine for supporting stable, risk-aware, and adaptive access control decisions. The engine maps risk_scores into allow, challenge, and block decisions using predefined thresholds and hysteresis margins.

The Zero Trust Policy Engine maps risk_scores into allow, challenge, and block decisions based on thresholds and hysteresis margins. By applying a small margin that limits the block rate to a minimum (m=2), the decision change rate between events drops from 0.152728 to 0.145744, while the block

rate remains close to the current configuration (0.132353 → 0.130309). A summary of the hysteresis effects is shown in Table 9.

Table 9 The effect of hysteresis (m=2, constrained) on decision stability.

Metric	No hysteresis (m=0)	With hysteresis (m=2)
Decision change rate	0.152728	0.145744
Escalation rate	0.075228	0.071907
De-escalation rate	0.077500	0.073837
Block rate	0.132353	0.130309

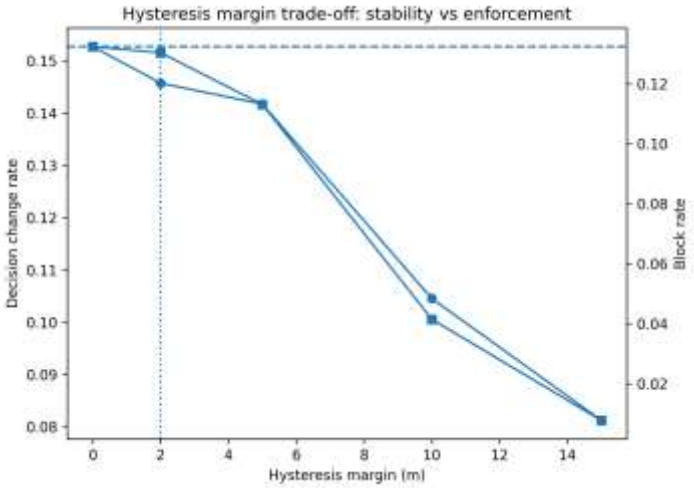


Fig. 5. Hysteresis margin trade-off: decision stability vs. enforcement strength.

Figure 5 shows a clear trade-off: the larger the hysteresis margin m, the lower the decision change rate (more stable decisions), but the block rate also decreases (the mitigation posture becomes more permissive). Therefore, the configuration used in this evaluation is m=2 (constrained), which lowers the change rate without significantly changing the block rate compared to the current policy.

The distribution of decisions in the policy engine (Table 10) shows that the majority of events are in the challenge category (267,632; 59.474%), followed by allow (122,809; 27.291%) and block (59,559; 13.235%). This composition represents a policy posture that holds most medium-risk events at the challenge stage before escalating to block.

Table 10 Distribution of Zero Trust policy engine decisions (dataset3).

Decision	Count	Percentage
allow	122,809	27.291%
challenge	267,632	59.474%

Decision	Count	Percentage
block	59,559	13.235%

The trade-off between detection and user friction is analyzed through threshold sweeps on the risk_score (Table 9). Example configurations show that a very aggressive threshold (thr=53) yields a recall of 1,000 but increases the predicted block rate to 0.723 (over-blocking). Conversely, a conservative threshold (thr=80; corresponding to a policy block rate of 0.132) suppresses friction but drops the recall on proxy blocks to 0.011. A moderate threshold (thr=69) approaches the WAF block rate (≈0.214) but still achieves low precision/recall (≈0.097), consistent with the limited performance of the risk_score rankings in Table 11.

Table 11 Summary of threshold sweep for block prediction (ŷ=1 if risk_score ≥ threshold).

Threshold	Precision	Recall	Pred. block rate
53	0.295815	1,000,000	0.723004
69	0.097136	0.097315	0.214271
80	0.018251	0.011294	0.132353

These findings demonstrate that ACF not only detects attack patterns, but also provides an adaptive decision-making mechanism that can be calibrated according to an organization's risk appetite.

4.4 Implications and Communication of Results

The results of this study not only demonstrate the technical performance of the developed artifact but also indicate broader contributions in academic and practical contexts. From a Design Science Research (DSR) perspective, these findings represent the communication stage of the results, where the proposed Adaptive Cloud Framework (ACF) artifact can serve as a conceptual and implementation reference in the development of behavior-based cloud security systems.

Academically, this research adds to the literature by integrating behavioral clustering and risk-based security decision-making approaches into a single adaptive framework. Practically, the results demonstrate that the behavioral-based approach can improve detection sensitivity to dynamic and previously undefined attack patterns. Therefore, the proposed artifact has the potential for adoption and further development in various cloud environments.

4.5 Discussion

4.5.1 Vulnerability of the Academic Journal Platform Ecosystem

The results of seed intent and behavioral clustering indicate that the majority of anomalous activity is not targeted sophisticated attacks, but rather automated reconnaissance exploiting uniform endpoint structures such *as /wp-content/, /wp-login.php, /index.php/index,* and OJS plugin paths.

This finding indicates that vulnerabilities stem not solely from individual domain weaknesses, but from the homogeneity of application architectures across institutions. Similar patterns of automated web browsing and predictable path exploitation have been widely reported in web security studies, particularly in CMS-based environments (Buczak & Guven, 2016; Sommer & Paxson, 2010).

In contrast to previous Zero Trust research which focused on user context-based identity and access control (Chandramouli & Butcher, 2023). This study shows that in the web-based scientific publication ecosystem, the dominant threat vector comes from automated web probing that exploits deterministic path patterns. Thus, security challenges in the academic cloud are not only identity-centric but also structure-centric.

The implication is that individual domain-based security approaches are insufficient. A model is needed that can understand collective, cross-domain dynamics, as facilitated by Ecosystem Analytics in ACF.

4.5.2 Conceptual Separation between Detection Score and Policy Score

Evaluation in Table 6 shows that anomaly_risk has a higher discriminatory ability (*AUROC 0.7296) than risk_score (0.5727*). This finding is architecturally important.

Previous studies combine detection and decision making into a single score (Ahmed et al., 2016; Chandola et al., 2009). However, the empirical results of this study indicate that the detection score (*anomaly_risk*) is optimal for ranking and prioritizing threats, and the policy score (*risk_score*) functions as a friction calibration mechanism (policy gating).

This separation demonstrates that detection and enforcement have different optimization goals. Detection optimizes statistical separability, while policy engines optimize the trade-off between mitigation and user friction. This finding is consistent with previous observations that anomaly detection systems often suffer from high false positive rates when used directly for enforcement decisions (Sommer & Paxson, 2010). The architectural contribution of this research lies in the explicit separation of these two functions within a single adaptive framework, which is rarely discussed in the Zero Trust or ML-based intrusion detection literature.

4.5.3 Cross-Domain Propagation as an Ecosystem Risk

Cross-domain propagation analysis (Table 3 and Figure 2) shows the presence of sources (IP hashes) that systematically interact with multiple domains within a limited timeframe. This indicates tool reuse and the exploitation of configuration uniformity. Most intrusion detection research evaluates performance on a single domain or a single isolated network (Ahmed et al., 2016). This study shows that in the academic cloud context, risks are ecosystem-wide, not local. Similar multi-vector and distributed attack behavior has been identified in recent cloud security studies (Lysenko et al., 2022; Neuschmied et al., 2022; Shieh et al., 2022).

Consequently, partial hardening on one domain is not enough, a cross-domain baseline security posture is required and monitoring must be centralized and pattern propagation-based.

Thus, ACF expands the security paradigm from domain-centric defense to ecosystem-aware adaptive defense.

4.5.4 Policy Stability through Hysteresis

Hysteresis evaluation (Table 7) shows that a small margin (m=2) is able to reduce the decision change rate without significantly changing the block rate. These findings have important practical implications. In Zero Trust implementations, decision oscillations can degrade usability and user trust. With hysteresis, ACF can stabilize enforcement without compromising mitigation posture. Although most Zero Trust research focuses primarily on authentication, access control, and policy strictness (Rose et al., 2020). The issue of decision stability is rarely discussed. This study introduces hysteresis as a mechanism to manage threshold sensitivity, thereby improving consistency in policy enforcement under dynamic risk conditions.

This concept shows that policy stability is a dimension that is rarely evaluated in Zero Trust research, which generally only focuses on detection accuracy.

4.5.5 The Role of Blockchain as an Audit Layer, not a Detection Layer

In this study, blockchain is positioned not as a detection component, but as an immutable audit layer. Testbed results demonstrate low latency (median 11 ms) and no hash duplication. Previous research has extensively explored the use of blockchain for data integrity, secure record-keeping, and decentralized trust (Dorri et al., 2017; Zyskind et al., 2015). However, many studies place blockchain as the primary security mechanism, which often leads to overly complex solutions.

Unlike many studies that claim blockchain as the primary security solution, this research positions blockchain in a minimalist and functional way, specifically to ensure non-repudiation and policy accountability. This approach avoids over-engineering and clarifies the functional boundaries between components: machine learning-based detection, Zero Trust enforcement, and blockchain-enabled accountability. This separation of roles strengthens the architecture's modularity while increasing implementation scalability.

4.5.6 Blockchain Audit Layer Performance Analysis

Table 12 demonstrated that smart contract-based audit logging on the testbed yielded a 100% success rate with no event_hash or tx_hash duplications. This indicates consistent logging integrity and the absence of collisions in the hashing mechanism used, which is consistent with previous findings on blockchain reliability in secure logging systems (Dorri et al., 2017).

Table 12 Summary of blockchain audit logging metrics on the testbed.

Metric	**Value**
Total transactions	120
Success rate (status=1)	1,000,000
Unique event_hash	120 (duplicates: 0)
Tx latency p50/p95/p99 (ms)	11.0 / 21.05 / 28.24

Metric	Value
Tx latency mean (ms)	12,308
Gas used mean / p50	23490.5 / 23492.0
Gas cost mean / p50 (wei)	2.454234e+13 / 2.349767e+13

However, the interpretation of these results must be placed in an architectural context. The blockchain component in ACF is not on the critical path for detection or enforcement, as the logging mechanism is implemented after the policy_decision is determined. Therefore, transaction latency does not directly impact real-time decisions in Zero Trust.

These findings reinforce the modular approach in ACF, where machine learning is used for detection, the Zero Trust Policy Engine for enforcement, and blockchain for accountability. This separation of functions allows for increased scalability, as the audit layer can be optimized or batch processed without disrupting the detection or decision-making process.

While the testbed results show promising performance, testing in high-throughput production scenarios is still needed to evaluate its impact on node resource consumption, transaction overhead, and the stability of the permissioned network under burst traffic conditions. Thus, the results in Table 10 provide initial evidence that integrating blockchain as an audit layer in ACF is feasible and does not incur significant overhead. However, validation of long-term scalability is still needed.

4.6 Limitations and Future Research

This study used proxy (block) labels and did not verify exploit success or post-incident impact. Furthermore, some numeric columns were all-NaN in a subset of the data and therefore did not contribute to the model. Cross-domain generalization needs to be tested more rigorously with more aggressive holdouts (e.g., entirely new domains and different time periods) and real-world incident-based evaluations. Valuable follow-up work includes: (i) sequential modeling (e.g., HMM/Transformer) for multi-step exploit chain patterns, (ii) enrichment with user-agent, ASN/geolocation (privacy-preserving) features, and (iii) validation with incident response/forensic data to link intent to impact. The blockchain component in this study is positioned as a hash-based audit log; comprehensive testing of transaction latency, throughput, and operational costs on a permissioned network is warranted as further work to ensure scalability to large-scale campus traffic.

5 Conclusion

This research demonstrates that web-based academic journal platforms, particularly in heterogeneous and distributed cloud ecosystems, face systemic and recurring threat patterns. Seed intent and behavioral clustering analyses reveal the dominance of automated web probing that exploits the uniformity of endpoint structures across domains. These findings confirm that security risks in academic clouds are not only domain-centric but also ecosystem-centric.

Through the integration of seed intent taxonomy, behavioral embedding, ecosystem analytics, unsupervised anomaly detection-based risk scoring, Zero Trust Policy Engine with hysteresis

mechanism, and blockchain-based immutable audit logging, this study proposes the Adaptive Cybersecurity Framework (ACF) as an ecosystem-based adaptive security model.

Theoretically, this research makes a major contribution in expanding the security paradigm from domain-level defense to ecosystem-aware adaptive defense. Furthermore, this research emphasizes the explicit separation between detection-oriented anomaly scoring and policy-oriented risk scoring in the context of Zero Trust. Another contribution lies in the integration of a feedback loop that connects detection, enforcement, and auditability processes in a single, modular, adaptive cycle.

Empirically, a large-scale production log-based evaluation demonstrated that *anomaly_risk* has sufficient discriminatory capacity for threat prioritization, while *risk_score* serves as a policy calibration mechanism based on an organization's risk appetite. The implementation of hysteresis was shown to improve decision stability without significantly changing the mitigation posture. Furthermore, testing of the blockchain audit layer demonstrated that immutable logging can be implemented with low latency overhead in a controlled testbed environment.

From a practical perspective, the results of this study provide an implementable blueprint for higher education institutions in designing security systems that are not only reactive to incidents but also adaptive to the dynamics of cross-domain attack patterns. This approach is relevant for strengthening the security of the increasingly digitalized scientific publication ecosystem.

However, this study has several limitations. The quantitative evaluation used WAF-based proxy labels and did not verify the success of actual exploits. Furthermore, the blockchain testing was conducted at a testbed scale and does not reflect high-scale production loads. Further research is needed to test new cross-domain generalizations, model multi-step attack chains, and evaluate blockchain performance on permissioned networks with higher throughput.

Overall, the Adaptive Cybersecurity Framework (ACF) offers a conceptual and empirical foundation for the development of an adaptive, scalable, and accountable academic cloud security system, while opening up further research on the integration of Machine Learning, Zero Trust, and Blockchain in the context of higher education digital ecosystem security.

Author Contributions:

Conceptualization	: DD. and IS
Methodology	: DD and TWY
Software	: DD
Validation	: DD, TWY, IS and TWL
Formal analysis	: DD and IS
Investigation	: DD and TWL
Resources	: DD
Data curation	: DD
Writing—original draft preparation	: DD

Writing—review and editing : TWY, IS and TWL

Visualization : TWL

Supervision : IS., TWY

Project administration : DD

Funding acquisition : DD

All authors have read and agreed to the published version of the manuscript

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Abbreviations

The following abbreviations are used in this manuscript:

ACF : Adaptive Cybersecurity Framework

MC : Machine Learning

LMS : Learning Management Systems

OJS : Open Journal System

ZTA : Zero Trust Architecture

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